

6 (long)

The cross-section of a composite beam built-up from a pair of timber beams and a concrete slab is shown in Fig. 6. The concrete and the timber are connected at the interface by a strong, stiff adhesive that allows fully composite structural behaviour to be developed. The dimensions of the beam are given in Fig. 6.

The Young's moduli of the concrete and the timber are 30 GPa and 10 GPa respectively. The maximum longitudinal stress that the concrete can sustain is 30 MPa in compression and zero MPa in tension. The maximum longitudinal stress that the timber can sustain is 20 MPa in both tension and compression. The composite beam is designed to be simply supported and resist gravity loads only, meaning that the top fibre will be in compression and the bottom fibre will be in tension.

- (a) Calculate the depth of the neutral axis, showing that this passes through the timber. Comment on the significance of the neutral axis passing through the timber rather than the concrete. [10]
- (b) Determine the flexural rigidity EI of the composite beam. [10]
- (c) Determine the maximum bending moment that the cross section can sustain. [10]

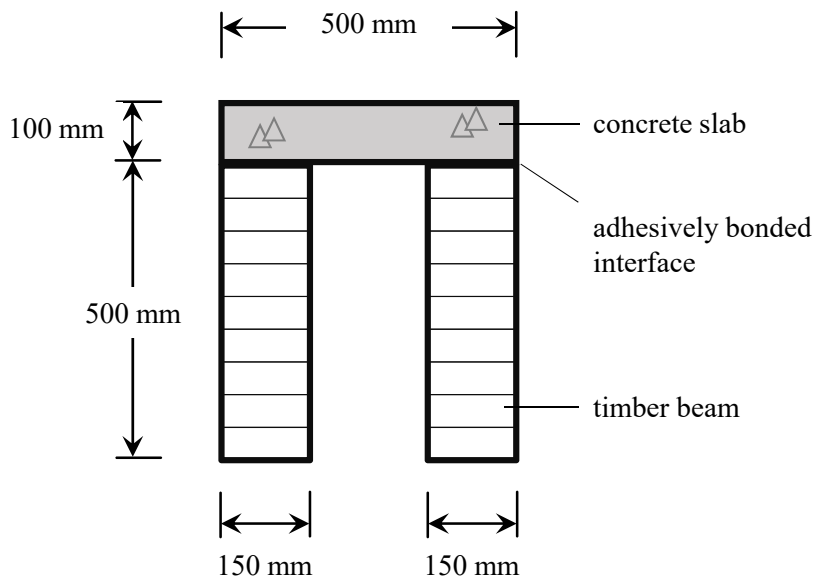


Fig. 6